Cambridge Assessment International Education

# Cambridge International AS & A Level

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*0296910415*

**MATHEMATICS** **9709/12**

Paper 1 Pure Mathematics 1 **October/November 2023**

**1 hour 50 minutes**

You must answer on the question paper.

You will need: List of formulae (MF19)

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

**INFORMATION**

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages.


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**1** The coefficient of $x^3$ in the expansion of $(3 + 2ax)^5$ is six times the coefficient of $x^2$ in the expansion of $(2 + ax)^6$.

Find the value of the constant $a$. [4]

**2** Find the exact solution of the equation

$$\tfrac{1}{6}\pi + \tan^{-1}(4x) = -\cos^{-1}\left(\tfrac{1}{2}\sqrt{3}\right). \qquad [2]$$

**3** The equation of a curve is such that $\frac{dy}{dx} = \frac{1}{2}x + \frac{72}{x^4}$. The curve passes through the point $P(2, 8)$.

**(a)** Find the equation of the normal to the curve at $P$. [2]

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**(b)** Find the equation of the curve. [4]

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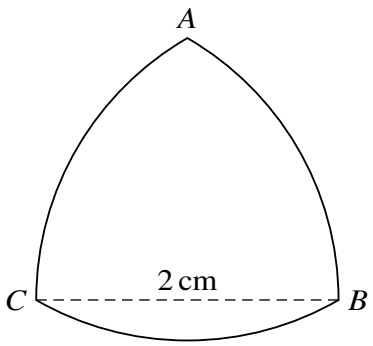
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**4**




The diagram shows the shape of a coin. The three arcs $AB$, $BC$ and $CA$ are parts of circles with centres $C$, $A$ and $B$ respectively. $ABC$ is an equilateral triangle with sides of length 2 cm.

**(a)** Find the perimeter of the coin. [2]

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**(b)** Find the area of the face $ABC$ of the coin, giving the answer in terms of $\pi$ and $\sqrt{3}$. [4]

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**5** The first, second and third terms of a geometric progression are $\sin\theta$, $\cos\theta$ and $2 - \sin\theta$ respectively, where $\theta$ radians is an acute angle.

**(a)** Find the value of $\theta$. [3]

**(b)** Using this value of $\theta$, find the sum of the first 10 terms of the progression. Give the answer in the form $\frac{b}{\sqrt{c}-1}$, where $b$ and $c$ are integers to be found. [3]

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**6** The equation of a curve is $y = x^2 - 8x + 5$.

**(a)** Find the coordinates of the minimum point of the curve. [2]

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The curve is stretched by a factor of 2 parallel to the $y$-axis and then translated by $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$.

**(b)** Find the coordinates of the minimum point of the transformed curve. [2]

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**(c)** Find the equation of the transformed curve. Give the answer in the form $y = ax^2 + bx + c$, where $a$, $b$ and $c$ are integers to be found. [4]

**7** **(a)** Verify the identity $(2x - 1)(4x^2 + 2x - 1) \equiv 8x^3 - 4x + 1$. [1]

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**(b)** Prove the identity $\dfrac{\tan^2\theta + 1}{\tan^2\theta - 1} \equiv \dfrac{1}{1 - 2\cos^2\theta}$. [3]

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**(c)** Using the results of **(a)** and **(b)**, solve the equation

$$\frac{\tan^2\theta + 1}{\tan^2\theta - 1} = 4\cos\theta,$$

for $0° \leqslant \theta \leqslant 180°$. [5]

**8** Functions f and g are defined by

$$\mathrm{f}(x) = (x + a)^2 - a \text{ for } x \leqslant -a,$$
$$\mathrm{g}(x) = 2x - 1 \text{ for } x \in \mathbb{R},$$

where $a$ is a positive constant.

**(a)** Find an expression for $\mathrm{f}^{-1}(x)$. [3]

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**(b)** **(i)** State the domain of the function $\mathrm{f}^{-1}$. [1]

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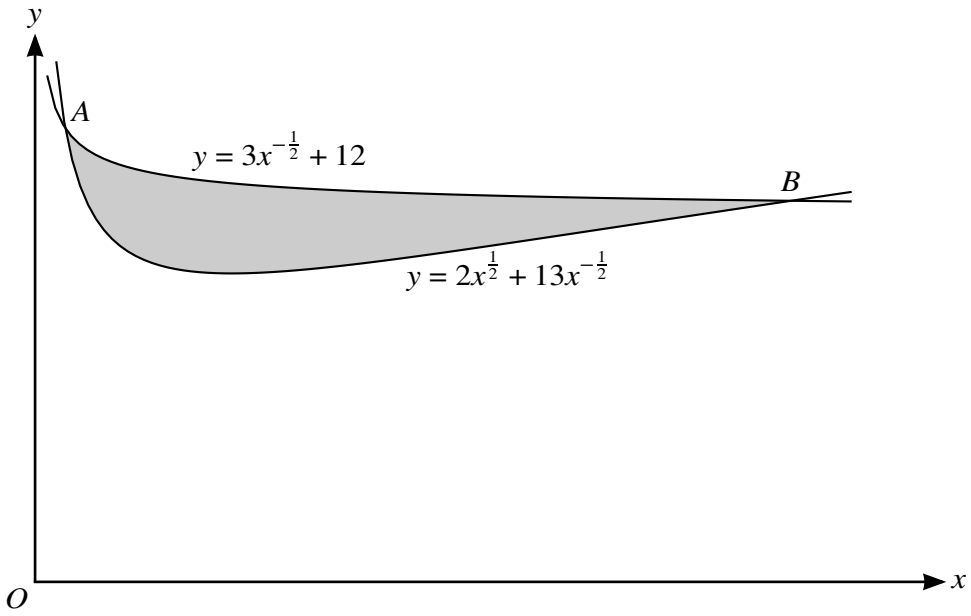
**(ii)** State the range of the function $\mathrm{f}^{-1}$. [1]

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**(c)** Given that $a = \frac{7}{2}$, solve the equation $\mathrm{gf}(x) = 0$. [3]

**9**




The diagram shows curves with equations $y = 2x^{\frac{1}{2}} + 13x^{-\frac{1}{2}}$ and $y = 3x^{-\frac{1}{2}} + 12$. The curves intersect at points $A$ and $B$.

**(a)** Find the coordinates of $A$ and $B$. [4]

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**(b)** Hence find the area of the shaded region. [5]

**10** The equation of a curve is $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = (4x - 3)^{\frac{5}{3}} - \frac{20}{3}x$.

**(a)** Find the $x$-coordinates of the stationary points of the curve and determine their nature. [6]

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**(b)** State the set of values for which the function f is increasing. [1]

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**11** The coordinates of points $A$, $B$ and $C$ are $(6, 4)$, $(p, 7)$ and $(14, 18)$ respectively, where $p$ is a constant. The line $AB$ is perpendicular to the line $BC$.

**(a)** Given that $p < 10$, find the value of $p$. [4]

A circle passes through the points $A$, $B$ and $C$.

**(b)** Find the equation of the circle. [3]

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**(c)** Find the equation of the tangent to the circle at $C$, giving the answer in the form $dx + ey + f = 0$, where $d$, $e$ and $f$ are integers. [3]

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## Additional Page

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